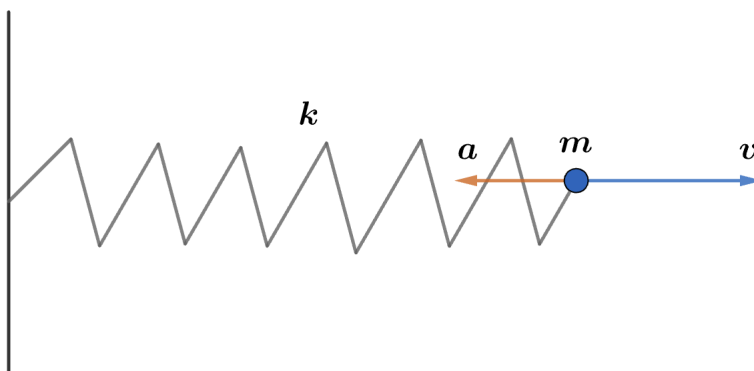


2022B F=ma Exam: Problem 23

Kevin S. Huang



By energy conservation,

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

For simple harmonic motion,

$$a = -\omega^2 x$$
$$E = \frac{1}{2}mv^2 + \frac{1}{2}\frac{k}{\omega^4}a^2$$

Thus,

$$\frac{v^2}{2E/m} + \frac{a^2}{2E\omega^4/k} = 1$$

is the equation of an ellipse in the variables v and a . The velocities and accelerations can take all signs so the answer is C.